Disk and annulus rigidity for the (h_1, h_1, U) problem

1 Introduction

Let $U \subset \mathbb{C}$ be a bounded, connected open set with C^1 boundary, and let $\partial U = \Gamma_0 \cup \Gamma_1 \cup \cdots \cup \Gamma_N$, where Γ_0 is the outer boundary component and $\Gamma_1, \ldots, \Gamma_N$ bound the holes $H_1, \ldots, H_N \subset \mathbb{C} \setminus \overline{U}$.

We consider the (h_1, h_1, U) localization problem: find all such U for which the first Hermite function h_1 is an eigenfunction of the h_1-window localization operator $\mathcal{L}_U^{h_1}$, i.e.,

$$\mathcal{L}_U^{h_1} h_1 = \lambda h_1$$

for some $\lambda \in \mathbb{R}$.

The main result of this note is the following.

Theorem 1.1. *Let $U \subset \mathbb{C}$ be bounded and connected with C^1 boundary, and suppose that $\mathcal{L}_U^{h_1} h_1 = \lambda h_1$ for some $\lambda \in \mathbb{R}$. Then U is a disk or an annulus centered at the origin.*

The proof combines the Bargmann-space formulation and the first-order distributional equation established in the companion note `h_1-loc.tex` with a new explicit-antiderivative argument that determines the solution in each connected component of $\mathbb{C}$.

2 Setup and distributional equation

We recall the relevant results from `h_1-loc.tex`. Under the Bargmann transform, $h_1 \leftrightarrow e_1(w) = \sqrt{\pi}\, w$, and a standard computation gives

$$\langle e_1, T_z e_1 \rangle_{\mathcal{F}^2} = e^{-\frac{\pi}{2}|z|^2}(1 - \pi|z|^2).$$

The eigenvalue equation $\mathcal{L}_U^{h_1} h_1 = \lambda h_1$ is therefore equivalent to

$$\int_U (1 - \pi|z|^2)(w - z)\, e^{\pi w \bar{z} - \pi|z|^2}\, dA(z) = \lambda w, \qquad w \in \mathbb{C}. \tag{1}$$

Set

$$g(w) := (1 - \pi|w|^2)\, e^{-\pi|w|^2}.$$

The Bargmann identity (1) is equivalent to the vanishing $\langle \omega, e^{\pi w z} \rangle = 0$ for all $w \in \mathbb{C}$, where

$$\omega := \big(-\tfrac{1}{\pi}\partial_w - \bar{w}\big)(g\, \mathbf{1}_U) + \tfrac{\lambda}{\pi}\, \partial_w \delta_0 \tag{2}$$

is a compactly supported distribution. (See `h_1-loc.tex`, Proposition 3.1 and its proof.) Expanding the Leibniz rule gives the equivalent form

$$\omega = \bar{w}\, e^{-\pi|w|^2} \mathbf{1}_U - \tfrac{1}{\pi}\, g(w)\, \partial_w \mathbf{1}_U + \tfrac{\lambda}{\pi}\, \partial_w \delta_0. \tag{3}$$

Since the exponentials $\{e^{\pi w z} : w \in \mathbb{C}\}$ span the kernel of ∂_w acting on distributions, the Ehrenpreis–Malgrange theorem furnishes a (unique) distribution $u \in \mathcal{E}'(\mathbb{C})$ satisfying

$$\partial_w u = \omega. \tag{4}$$

Lemma 2.1. *The solution of* (4) *in $\mathcal{E}'(\mathbb{C})$ is unique.*

Proof. If $T \in \mathcal{E}'(\mathbb{C})$ satisfies $\partial_w T = 0$, then T is anti-holomorphic as a distribution. Its Fourier transform satisfies $(\xi_1 - i\xi_2)\hat{T}(\xi_1, \xi_2) = 0$, so $\hat{T}$ is supported at the origin, hence T is a polynomial. A compactly supported polynomial vanishes identically. $\square$

3 Regularity and structure of the solution

We now analyze the unique solution $u \in \mathcal{E}'(\mathbb{C})$ of (4) region by region.

3.1 Unbounded exterior

In the unbounded component Ω_∞ of $\mathbb{C} \setminus \overline{U}$, both $\mathbf{1}_U$ and δ_0 vanish, so $\partial_w u = 0$. Thus u is anti-holomorphic in Ω_∞. Since u has compact support, $u \equiv 0$ in Ω_∞ (an anti-holomorphic function that vanishes on an open subset of a connected set vanishes identically).

3.2 Interior of U

In U, the boundary distribution $\partial_w \mathbf{1}_U$ vanishes, and (3) reduces to

$$\partial_w u = \bar{w}\, e^{-\pi|w|^2} + \tfrac{\lambda}{\pi}\, \partial_w \delta_0 \qquad \text{in } \mathcal{D}'(U). \tag{5}$$

Define $v := u - \frac{\lambda}{\pi}\, \delta_0 \in \mathcal{D}'(U)$. Then $\partial_w v = \bar{w}\, e^{-\pi|w|^2}$, which is C^∞. Since ∂_w is elliptic (its principal symbol $\xi_1 - i\xi_2$ is nonvanishing for $\xi \neq 0$), elliptic regularity gives $v \in C^\infty(U)$.

Lemma 3.1. *The function v is real-valued in U.*

Proof. Applying $\partial_{\bar{w}}$ to $\partial_w v = \bar{w}\, e^{-\pi|w|^2}$:

$$\tfrac{1}{4}\Delta v = \partial_{\bar{w}}(\bar{w}\, e^{-\pi|w|^2}) = e^{-\pi|w|^2} + \bar{w} \cdot (-\pi w)\, e^{-\pi|w|^2} = (1 - \pi|w|^2)\, e^{-\pi|w|^2}.$$

This is real, so $\Delta(\operatorname{Im} v) = 0$ in U; i.e., $\operatorname{Im} v$ is harmonic in U.

We will show that $\operatorname{Im} v = 0$ on ∂U in the distributional trace sense, after which the maximum principle gives $\operatorname{Im} v \equiv 0$. The boundary analysis is carried out in §4 below. □

3.3 Holes not containing the origin

Let H_j $(1 \le j \le N)$ be a hole with $0 \notin H_j$. In H_j, both $\mathbf{1}_U$ and δ_0 vanish, so $\partial_w u = 0$. Hence u is anti-holomorphic in H_j.

3.4 The hole containing the origin (if any)

If $0 \in U$, there is nothing to add. If $0 \notin U$, then 0 lies in some hole H_0 (relabeling if necessary). In H_0,

$$\partial_w u = \tfrac{\lambda}{\pi}\, \partial_w \delta_0 \qquad \text{in } \mathcal{D}'(H_0).$$

Setting $\tilde{u} := u - \frac{\lambda}{\pi}\, \delta_0$, we get $\partial_w \tilde{u} = 0$ in $\mathcal{D}'(H_0)$, so $\tilde{u}$ is anti-holomorphic (hence C^∞) in H_0. In particular, u is represented by a smooth function away from 0 in H_0, with the distributional singularity $\frac{\lambda}{\pi}\delta_0$ at the origin.

4 Boundary traces and the jump relation

Since $u \in \mathcal{E}'(\mathbb{C})$ satisfies $\partial_w u = \omega$, we can read off the jump of u across ∂U by comparing the coefficients of the boundary distribution $\partial_w \mathbf{1}_U = -\frac{1}{2}(\nu_1 - i\nu_2)\, \mathcal{H}^1|_{\partial U}$.

Write $u = v\, \mathbf{1}_U + \sum_j u_j\, \mathbf{1}_{H_j} + 0 \cdot \mathbf{1}_{\Omega_\infty}$ distributionally near ∂U (ignoring the δ_0 term, which is supported away from ∂U). Then

$$\partial_w u = (\partial_w v)\, \mathbf{1}_U + \sum_j (\partial_w u_j)\, \mathbf{1}_{H_j} + \big(v|_{\partial U} - u_{\text{ext}}|_{\partial U}\big) \partial_w \mathbf{1}_U + \cdots,$$

where u_{ext} denotes the exterior trace (which is 0 on Γ_0 and u_j on Γ_j). Matching the coefficient of $\partial_w \mathbf{1}_U$ in (3):

$$v|_{\Gamma_k} - u_{\text{ext}}|_{\Gamma_k} = -\tfrac{1}{\pi}\, g(w) = -\tfrac{1}{\pi}(1-\pi|w|^2)\, e^{-\pi|w|^2} \qquad (k = 0, 1, \ldots, N). \tag{6}$$

On the outer boundary Γ_0: $u_{\text{ext}} = 0$, so

$$v|_{\Gamma_0} = -\tfrac{1}{\pi}(1-\pi|w|^2)\, e^{-\pi|w|^2} \qquad \text{(real)}. \tag{7}$$

On an inner boundary Γ_j: $u_{\text{ext}} = u_j$, so

$$v|_{\Gamma_j} - u_j|_{\Gamma_j} = -\tfrac{1}{\pi}(1-\pi|w|^2)\, e^{-\pi|w|^2} \qquad \text{(real)}. \tag{8}$$

5 Reality of u everywhere

Proposition 5.1. *The smooth function v is real-valued in U, and the anti-holomorphic function u_j is real-valued in each hole H_j.*

Proof. Define the function $\Phi : \mathbb{C} \to \mathbb{R}$ by

$$\Phi(w) := \begin{cases} \operatorname{Im} v(w) & w \in U, \\ \operatorname{Im} u_j(w) & w \in H_j \ (j = 1, \ldots, N), \\ 0 & w \in \Omega_\infty. \end{cases}$$

In each region, Φ is harmonic:

- In U: $\Delta(\operatorname{Im} v) = \operatorname{Im}(\Delta v) = 0$ since $\Delta v = 4(1-\pi|w|^2)e^{-\pi|w|^2}$ is real (Lemma 3.1).
- In H_j: u_j is anti-holomorphic, hence harmonic, so $\operatorname{Im} u_j$ is harmonic.
- In Ω_∞: $\Phi \equiv 0$.

Moreover, Φ is continuous across ∂U. Indeed:

- On Γ_0: from the U-side, $\Phi = \operatorname{Im} v$; from the exterior, $\Phi = 0$. The jump relation (7) gives $v|_{\Gamma_0} = \text{real}$, so $\operatorname{Im} v|_{\Gamma_0} = 0$. Thus Φ is continuous across Γ_0.
- On Γ_j: the jump relation (8) gives $v|_{\Gamma_j} - u_j|_{\Gamma_j} = \text{real}$, so $\operatorname{Im} v|_{\Gamma_j} = \operatorname{Im} u_j|_{\Gamma_j}$. Thus Φ is continuous across Γ_j.

Therefore Φ is harmonic in $U \cup H_1 \cup \cdots \cup H_N \cup \Omega_\infty = \mathbb{C}$ (since it is harmonic in each piece and continuous across the interfaces, it is harmonic globally by removability of C^1 interfaces for continuous harmonic functions). Since $\Phi \equiv 0$ outside a compact set, the maximum principle (or Liouville's theorem) gives $\Phi \equiv 0$ on $\mathbb{C}$.

In particular, $\operatorname{Im} v \equiv 0$ in U and $\operatorname{Im} u_j \equiv 0$ in each H_j. □

Remark 5.2. *For the hole H_0 containing the origin (if $0 \notin U$): we showed $u = \tilde{u} + \frac{\lambda}{\pi}\delta_0$ with $\tilde{u}$ anti-holomorphic in H_0. The jump relation gives $\tilde{u}|_{\Gamma_0^{H_0}}$ real (since v and g are real on the boundary). The same maximum-principle argument shows $\operatorname{Im} \tilde{u} \equiv 0$, so $\tilde{u}$ is real in H_0.*

6 Explicit formula and rigidity

Proposition 6.1. *The solution satisfies*

$$v(w) = -\frac{1}{\pi}\, e^{-\pi|w|^2} + C \qquad (w \in U) \tag{9}$$

for a constant $C \in \mathbb{R}$. Moreover, each u_j is a real constant in H_j.

Proof. Since v is real and smooth in U, the first-order equation $\partial_w v = \bar{w}\, e^{-\pi|w|^2}$ gives, via $\partial_w = (\partial_x - i\partial_y)/2$,

$$\frac{\partial_x v - i\,\partial_y v}{2} = (x - iy)\, e^{-\pi(x^2+y^2)}.$$

Since v is real, matching real and imaginary parts:

$$\partial_x v = 2x\, e^{-\pi|w|^2}, \qquad \partial_y v = 2y\, e^{-\pi|w|^2}. \tag{10}$$

Now observe that the function $p(w) := -\frac{1}{\pi}\, e^{-\pi|w|^2}$ satisfies

$$\partial_x p = 2x\, e^{-\pi|w|^2}, \qquad \partial_y p = 2y\, e^{-\pi|w|^2}.$$

Therefore $\nabla(v - p) \equiv 0$ in U. Since U is connected, $v = p + C$ for some constant C. This proves (9).

For the holes: each u_j is real and anti-holomorphic in the connected open set H_j. Write $u_j = \varphi_j(\bar{w})$ for a holomorphic function φ_j. Since u_j is real, $\overline{\varphi_j(\bar{w})} = \varphi_j(\bar{w})$, i.e., $\overline{\varphi_j(\zeta)} = \varphi_j(\bar{\zeta})$ for $\zeta \in \overline{H_j}$. Expanding $\varphi_j(\zeta) = \sum_{n\geq 0} a_n(\zeta - \zeta_0)^n$ locally, the reality condition forces $a_n = 0$ for all $n \geq 1$ (since $\bar{a}_n e^{in\theta} = a_n e^{-in\theta}$ for all θ requires $a_n = 0$). Hence φ_j is constant, so $u_j \equiv c_j \in \mathbb{R}$. □

Proposition 6.2. *Every connected component of ∂U is a circle centered at the origin.*

Proof. **Outer boundary.** On Γ_0, the jump relation (7) and (9) give

$$-\frac{1}{\pi}\, e^{-\pi|w|^2} + C = -\frac{1}{\pi}(1 - \pi|w|^2)\, e^{-\pi|w|^2} = -\frac{1}{\pi}\, e^{-\pi|w|^2} + |w|^2\, e^{-\pi|w|^2}.$$

Hence

$$|w|^2\, e^{-\pi|w|^2} = C \qquad \text{on } \Gamma_0. \tag{11}$$

Inner boundaries. On Γ_j, the jump relation (8) gives

$$\Big(-\frac{1}{\pi}\, e^{-\pi|w|^2} + C\Big) - c_j = -\frac{1}{\pi}(1 - \pi|w|^2)\, e^{-\pi|w|^2},$$

so

$$|w|^2\, e^{-\pi|w|^2} = C - c_j \qquad \text{on } \Gamma_j. \tag{12}$$

Constancy of $|w|$. Consider the function $h : [0, \infty) \to \mathbb{R}$ defined by $h(t) := t\, e^{-\pi t}$. One has $h(0) = 0$, $h'(t) = (1 - \pi t)\, e^{-\pi t}$, so h is strictly increasing on $[0, \frac{1}{\pi}]$ and strictly decreasing on $[\frac{1}{\pi}, \infty)$, with maximum $h(\frac{1}{\pi}) = \frac{1}{\pi e}$.

On each boundary component Γ_k, equations (11)–(12) assert that $h(|w|^2) = \text{const}$. Since Γ_k is connected, $t(s) := |\gamma(s)|^2$ is a continuous function of the arclength parameter s. Its range is a connected subset of the level set $h^{-1}(\text{const})$. But this level set consists of at most two points (one in $[0, \frac{1}{\pi}]$ and one in $[\frac{1}{\pi}, \infty)$, if the constant is in the range of h). A connected subset of a two-point set is a single point. Therefore $|w|^2$ is constant on Γ_k, i.e., Γ_k is a circle centered at the origin. □

Proof of Theorem 1.1. By Proposition 6.2, every boundary component of U is a circle centered at the origin. Each hole H_j is therefore a disk $D(0, r_j)$. If there are two or more holes with radii $r_1 > r_2$, then $D(0, r_2) \subset D(0, r_1)$, so H_2 is contained in H_1, contradicting the fact that H_1 and H_2 are distinct connected components of $\mathbb{C} \setminus \overline{U}$. Thus there is at most one hole.

- If $N = 0$: $U = D(0, R)$ is a disk centered at the origin.
- If $N = 1$: $U = \{r < |w| < R\}$ is an annulus centered at the origin.

This completes the proof. □

7 The distributional trace: detailed justification

In this section we supply the technical details for the boundary analysis used above. The distributional equation $\partial_w u = \omega$ holds in $\mathcal{E}'(\mathbb{C})$. We must justify that the jump of u across ∂U is well-defined and given by (6), and that the function Φ from Proposition 5.1 is indeed continuous across ∂U.

Lemma 7.1. *The smooth function v extends continuously to $\overline{U}$, and the anti-holomorphic function u_j extends continuously to $\overline{H_j}$, for each j.*

Proof. By Proposition 6.1, $v = -\frac{1}{\pi}e^{-\pi|w|^2} + C$, which extends smoothly to all of $\mathbb{C}$. (One might object that Proposition 6.1 relies on Proposition 5.1, which uses Lemma 7.1. We resolve this circularity below.)

To avoid circularity, we argue as follows. In U, the general solution of $\partial_w v = \bar{w}\, e^{-\pi|w|^2}$ is

$$v(w) = -\frac{1}{\pi}\, e^{-\pi|w|^2} + C_0 + \varphi(\bar{w}),$$

where φ is anti-holomorphic in U and $C_0 \in \mathbb{C}$. The first term extends smoothly to $\overline{U}$. It remains to show that φ extends continuously to $\overline{U}$.

The distributional equation $\partial_w u = \omega$ forces the jump $v|_{\Gamma_k} - u_{\text{ext}}|_{\Gamma_k} = -\frac{1}{\pi}g$ as a distributional identity on Γ_k. Since the right-hand side and $-\frac{1}{\pi}e^{-\pi|w|^2} + C_0$ are both continuous on Γ_k, the distributional trace of φ on Γ_k (from the U-side) is a continuous function.

Now $\varphi(\bar{w})$ is harmonic in U. A harmonic function in a bounded domain with C^1 boundary whose distributional trace is continuous extends continuously to the closure (this follows from the barrier argument: C^1 domains satisfy the exterior cone condition at every boundary point, so the Wiener criterion gives continuity of the Perron solution, which must agree with φ by uniqueness).

The same argument applies to each u_j (anti-holomorphic, hence harmonic, with continuous distributional trace given by the jump relation). □

With boundary continuity established, the maximum-principle argument in Proposition 5.1 is rigorous: Φ is harmonic in each piece, continuous across boundaries, and vanishes at infinity, so $\Phi \equiv 0$ by the maximum principle for subharmonic functions on $\mathbb{C}$.

Once $\Phi \equiv 0$, Proposition 6.1 follows without circularity, and Proposition 6.2 and Theorem 1.1 follow in turn.

8 Remarks

Remark 8.1 (Why the argument is specific to the (h_1, h_1) case)**.** *The key algebraic input is that*

$$\partial_{\bar{w}}\big(\bar{w}\, e^{-\pi|w|^2}\big) = (1 - \pi|w|^2)\, e^{-\pi|w|^2}$$

is real-valued. *This is what makes Δv real, forcing* $\operatorname{Im} v$ *to be harmonic and ultimately v to be real.*

For the (h_0, h_0, U) problem, the interior first-order equation is $\partial_w u = -e^{-\pi|w|^2}$, and

$$\partial_{\bar{w}}\big(-e^{-\pi|w|^2}\big) = \pi w\, e^{-\pi|w|^2},$$

which is complex. *The Laplacian of the imaginary part is $4\pi y\, e^{-\pi|w|^2} \neq 0$, so the reality argument fails and the explicit antiderivative approach has no traction.*

Remark 8.2 (Comparison with the simply connected case)**.** *For simply connected U, the result was already accessible via two other routes:*

(i) *The angular-derivative argument applied to the second-order compact-support equation* $\Delta u = \rho\,\mathbf{1}_U - c\,\delta_0$ *(with* ρ *radial), which gives* u *radial and hence* U *a disk. This uses* $\nabla u = 0$ *on* ∂U*, which holds only for simply connected* U*.*

(ii) *The compatibility condition* $(1-\pi|w|^2)(w\cdot\tau)=0$ *from* `h_1-loc.tex`*, combined with analyticity of the boundary (if available from Kinderlehrer–Nirenberg).*

The present argument has the advantage of requiring neither the angular-derivative mechanism nor boundary analyticity, and of handling the multiply connected case directly.

Remark 8.3 (The admissible radii)**.** *For a disk* $U = D(0,R)$*, the constant* $C = R^2 e^{-\pi R^2}$ *and the eigenvalue* λ *are determined by the quadrature identity* (1)*. For an annulus* $U = \{r < |w| < R\}$*, the two boundary conditions give*

$$R^2 e^{-\pi R^2} = C, \qquad r^2 e^{-\pi r^2} = C - c_1,$$

and additionally the eigenvalue equation (1) *must hold, imposing a further relation among* r*,* R*,* λ*, and* c_1*. Whether these constraints admit solutions is a separate question; the present note only establishes the geometric rigidity.*

9 Validity Check

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I checked [h_1-disk-annulus.tex](/home/joao-pedro-ramos/deepseek/Inverse-Localize/Higher%20c

**Main Fatal Issue: Wrong First-Order Direction**

At [lines 92-96](/home/joao-pedro-ramos/deepseek/Inverse-Localize/Higher%20order%20Hermite%2

\[
\langle \omega,e^{\pi wz}\rangle=0
\]

implies existence of compactly supported \(u\) with

\[
\partial_w u=\omega.
\]

This is the wrong operator direction.

With the convention used later in the file,

\[
\partial_w=\frac12(\partial_x-i\partial_y)=\partial_z,
\]

the functions \(e^{\pi wz}\) are holomorphic in \(z\), so they lie in the kernel of \(\parti

For compactly supported solvability of

\[
\partial_z u=\omega,
\]
```

one needs \(\omega\) to annihilate anti-holomorphic entire test functions, for example \(e^{

\[
\partial_w u=\omega
\]

is not justified.

This matters because all later rigidity depends on the interior equation

\[
\partial_w v=\bar w e^{-\pi |w|^2},
\]

which then gives

\[
\frac14\Delta v=\partial_{\bar w}(\bar w e^{-\pi|w|^2})
=(1-\pi |w|^2)e^{-\pi|w|^2},
\]

a real right-hand side. If the valid equation is instead a \(\bar\partial\)-equation, this r

Second Fatal Issue: The Global Harmonicity Claim Is False

At [lines 263-268](/home/joao-pedro-ramos/deepseek/Inverse-Localize/Higher%20order%20Hermite

That is false. Smooth curves are not removable for harmonic functions under mere continuity.

A simple counterexample is

\[
\Phi(z)=
\begin{cases}
0, & |z|\ge 2,\\
\log(|z|/2), & 1<|z|<2,\\
\log(1/2), & |z|\le 1.
\end{cases}
\]

This function is continuous, compactly supported, harmonic in each region, and nonzero. Its

For this argument to work, one would need more than continuity of imaginary parts across \(\

Additional Technical Gaps

The jump comparison around [lines 184-205](/home/joao-pedro-ramos/deepseek/Inverse-Localize/

The boundary-regularity lemma at [lines 402-435](/home/joao-pedro-ramos/deepseek/Inverse-Loc

What Remains Sound

The algebra after the assumed equation

$$
\partial_w v=\bar w e^{-\pi|w|^2}
$$

is mostly correct. In particular, if one somehow had a real-valued v, then

$$
v(w)=-\frac1\pi e^{-\pi|w|^2}+C
$$

is correct, and the level-set argument using

$$
|w|^2 e^{-\pi |w|^2}=\text{constant}
$$

does force each boundary component to be a centered circle. The final geometric step from ce

But the proof does not validly reach the real-valuedness or the explicit formula.

Verdict

The proposed proof does **not** currently establish disk/annulus rigidity. The decisive brea